Comparative Analysis of Voltage Rise Mitigation Strategies for High-Penetration Solar PV Integration in Low-Voltage Distribution Networks

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***Abstract*—Rooftop solar photovoltaic (PV) integration in low-voltage (LV) residential distribution networks causes reverse power flow that elevates bus voltages beyond the EN 50160 statutory limit of 1.05 pu. This paper presents a systematic comparative study of six voltage rise mitigation strategies on the IEEE European LV Test Feeder—a 19-bus, 400 V, 200 kVA radial network with cable R/X = 3.1. The strategies evaluated are: (1) no mitigation baseline, (2) active power curtailment, (3) distributed electrical energy storage systems (EESS), (4) fixed power-factor control at PF = 0.95 lagging, (5) voltage-droop Q(U) control, and (6) a proposed adaptive droop whose gain scales linearly with PV penetration level. A forward-sweep load-flow framework is applied across penetration levels from 0% to 60%. Results show that curtailment and EESS are voltage-equivalent—both extend hosting capacity from 20% to 60%—yet EESS stores the 29.5 kW otherwise discarded at 60% penetration for evening dispatch. Among reactive power strategies, the proposed adaptive droop achieves V_{max} = 1.069 pu at 60% penetration versus 1.091 pu for fixed PF control, extending hosting capacity to 50% against 30% for fixed PF. Findings provide a quantitative decision framework for distribution operators selecting regulation-compliant PV integration strategies.**

***Index Terms*—Solar photovoltaic, voltage rise mitigation, low-voltage distribution network, EESS, reactive power control, droop control, hosting capacity, IEEE European LV test feeder.**

I. Introduction

The global installed capacity of solar photovoltaic (PV) generation has grown at an unprecedented pace, driven by declining technology costs, government incentives, and carbon neutrality targets [1], [9]. In residential LV distribution networks, rooftop PV installations now dominate distributed generation. However, these networks were designed for unidirectional power delivery; PV injection creates reverse power flows that progressively elevate bus voltages, potentially violating the 1.05 pu statutory limit of EN 50160 and IEEE Std 1547 [2].

This voltage rise is uniquely pronounced in LV networks due to their high resistance-to-reactance (R/X) ratio, typically exceeding 3. As established by Yang *et al.* [1], active power—not reactive power—is the primary voltage driver, making HV mitigation strategies significantly less effective at the LV level.

Published mitigation methods span two categories. Active power methods include PV curtailment [3] and distributed EESS [1], [4]. Reactive power methods include fixed power-factor control [5], voltage-droop Q(U) control [1], [6], and optimisation-based coordinated control [7]. Hybrid approaches have also been explored [8].

Three gaps motivate this work. First, direct side-by-side comparison under identical conditions remains sparse. Second, the energetic distinction between curtailment and EESS is rarely quantified in a single paper. Third, no study has evaluated a penetration-adaptive droop against its fixed-gain counterpart.

This paper makes four specific contributions:

- A unified forward-sweep load-flow simulation of six strategies on the IEEE European LV Test Feeder across 0–60% penetration.
- Quantification of the energy stored vs. wasted trade-off between EESS and curtailment under identical voltage constraints.
- Proposal and evaluation of an adaptive droop law, $K_q^{adapt} = K_q(1 + \alpha \cdot L_{PV})$, whose gain scales with PV penetration.
- A consolidated hosting-capacity decision framework for distribution system operators.

Section II reviews the voltage rise mechanism. Section III describes the test system. Section IV presents each strategy. Section V discusses results. Section VI concludes.

II. Voltage Rise in LV Distribution Networks

A. Governing Equations

For a two-bus feeder segment with impedance Z = R + jX carrying complex power P + jQ, the voltage magnitude deviation is approximated as [1]:

$$\left|\Delta\dot{V}\right| \approx \frac{PR+Q}{V_T} \tag{1}$$

The sensitivity of voltage magnitude to incremental power is:

$$d\left|\Delta\dot{V}\right| = \left(\frac{R}{V_T}\right) dP + \left(\frac{X}{V_T}\right) dQ \tag{2}$$

Key implication: With R/X ≈ 3–4 in LV cable feeders, active power drives voltage rise ≈3× more strongly than reactive power. Active power management is therefore the primary mitigation lever.

B. Forward-Sweep Load Flow

For a radial feeder with n buses and uniform net injection, the voltage at bus k is computed iteratively as:

$$V_{k+1} = V_k + \frac{P_{flow,k} \cdot R + Q_{flow,k} \cdot X}{V_k} \tag{3}$$

where the total net active power through segment k is:

$$P_{flow,k} = (P_{pv} - P_{load})(n - k + 1) \tag{4}$$

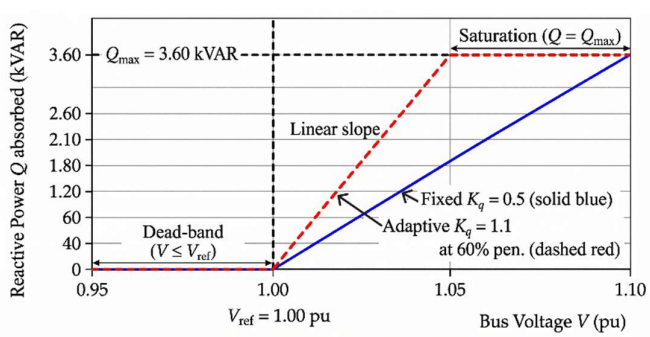
Injections accumulate bus-by-bus, making the terminal bus (Bus 19) the critical monitoring point.

C. Droop Control Theory

The voltage-droop Q(U) characteristic is a locally implemented, communication-free reactive power law. Each inverter absorbs reactive power proportional to the local voltage deviation above a reference V_{ref}:

$$Q_{inv(V)} = \begin{cases} -min(K_q(V - V_{ref}), Q_{max}) & if\ V > V_{ref} \\ 0 & if\ V \leq V_{ref} \end{cases} \tag{5}$$

here K_q is the droop gain (kVAR/pu), Q_{max} the inverter reactive power limit, and the negative sign denotes absorption (lagging). Fig. 1 illustrates the Q(U) characteristic for both fixed and adaptive droop gains.



Voltage-droop $Q(U)$ characteristic. Inverter absorbs Q only when $V > V_{ref} = 1.00$ pu.

Fig. 1. Voltage-droop Q(U) characteristic. Fixed droop (K_q = 0.5, solid blue) and adaptive droop (K_q = 1.1 at 60% penetration, dashed red). Saturation at Q_{max} = 3.60 kVAR per IEEE 1547-2018.

III. Test System and Simulation Methodology

A. IEEE European LV Test Feeder

All studies use the IEEE European LV Test Feeder [12], a 19-bus, 400 V (line-to-line), 200 kVA radial network representing a typical European residential feeder. It employs 95 mm² XLPE underground cables in 100 m segments with

R = 0.040 pu/segment ($R_{seg} = 0.032\ \Omega$)

X = 0.013 pu/segment ($X_{seg} = 0.011\ \Omega$)

R/X = 3.1.

The high R/X confirms active power management as the primary control lever. Fig. 2 shows the feeder topology with PV, load, and EESS at every bus.

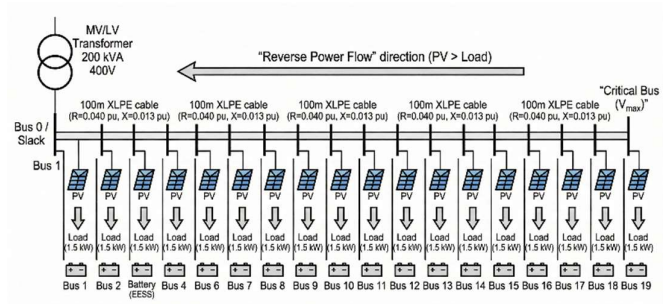


Fig. 2. IEEE European LV Test Feeder. The 19-bus, 400 V radial network supplies 19 residential loads (1.5 kW each). PV systems (up to 7.76 kWp) and home batteries (1.94 kW / 9.7 kWh) are co-located at each bus. Voltage rises from the transformer toward Bus 19 under reverse power flow.

B. Load and PV Parameters

Each bus represents one household: P_{load} = 1.50 kW, Q_{load} = 0.50 kVAR (PF = 0.95), and PV nameplate at 100% penetration $P_{pv}^{100\%}$ = 7.76 kWp. PV penetration is defined as:

$$L_{PV} = \frac{S_{PV,feeder}}{n_{loads} \cdot S_r} \times 100\% \tag{6}$$

where S_r = 7.76 kVA and n_{loads} = 19. At 60% penetration, net injection per bus is 3.16 kW into the feeder.

C. EESS Sizing

Home batteries are sized at 25% of PV nameplate power:

$$P_{batt,max} = 0.25 \times P_{pv}^{100\%} = 1.94\ kW \tag{7}$$

This corresponds to a 9.7 kWh battery at C/2 rate. At 60% penetration, the required charging power is 1.65 kW (85% of capacity), confirming the sizing is adequate.

D. Simulation Methodology

The forward-sweep algorithm of (3) is applied for each penetration level (0–60%, 10% steps) and each strategy. The transformer terminal is fixed at V_{tx} = 1.00 pu (slack). The voltage limit is V_{max} = 1.05 pu per EN 50160. For iterative strategies (3, 5, 6), convergence is declared when $\max|V^{i+1} - V^i| < 10^{-8}$ pu. Hosting capacity is the highest penetration at which all buses satisfy V ≤ V_{max} able I summarises all parameters.

TABLE I

Simulation Parameters

Parameter	Value	Source
n_{bus}	19	IEEE Euro. LV
V_{base}	400 V	IEC std.
S_{base}	200 kVA	IEEE Euro. LV
R (per segment)	0.040 pu	95 mm² XLPE
X (per segment)	0.013 pu	95 mm² XLPE
R/X	3.1	Calculated
P_{load} / bus	1.50 kW	Typical residential
P_{pv} (100%)	7.76 kWp	[1]

Parameter	Value	Source
V_{max}	1.05 pu	EN 50160
V_{ref}	1.00 pu	Nominal
$P_{batt,max}$	1.94 kW	25% of PV
PF_{static}	0.95 (lag)	[1]
K_q (fixed droop)	0.5 kVAR/pu	Tuned
$Q_{max,inv}$	3.60 kVAR	IEEE 1547-2018
α (adapt. droop)	2.0	Proposed

IV. Voltage Rise Mitigation Strategies

A. Strategy 1: No Mitigation (Baseline)

All inverters operate at unity PF with full active output . This establishes the worst-case voltage profile against which all other strategies are assessed. At 60% penetration, V_{Bus19} = 1.108 pu, exceeding V_{max} = 5.5%.

B. Strategy 2: Active Power Curtailment

Each inverter limits its output $P_{pv,c}$ to satisfy V ≤ V_{max} , permanently discarding ΔP = $P_{pv} - P_{pv,c}$:

$$\min_{\{P_{pv,c,i}\}} \sum_i P_{pv,c,i} \quad s.t. |V_k| \le V_{max}, \forall k \tag{8}$$

At 60% penetration, 29.5 kW of solar energy per hour is permanently discarded across the feeder.

C. Strategy 3: Distributed EESS

Home batteries charge at P_{batt} to absorb surplus PV:

$$P^*_{batt} = arg \min_{P_{batt}} P_{batt} \quad s.t. |V_k| \le V_{max}, 0 \le P_{batt} \le P_{batt,max} \tag{9}$$

Net injection becomes $P_{pv} - P_{load} - P_{batt}$. Unlike curtailment, all energy is stored for evening dispatch, preserving all PV generation.

D. Strategy 4: Static Reactive Power Control (Fixed PF)

Inverters permanently operate at PF = 0.95 lagging:

$$Q_{inv} = -P_{pv} \cdot tan\big(arccos(0.95)\big) = -0.329\, P_{pv} \tag{10}$$

This single-pass method absorbs Q regardless of voltage level, which is wasteful at low penetration and inherently limited by the high R/X ratio.

E. Strategy 5: Voltage-Droop Reactive Power Control

Inverters implement (5) with fixed gain K_q = 0.5 kVAR/pu. Reactive power is absorbed only when V > V_{ref} = 1.00 pu, eliminating wasteful absorption at low penetration. Reactive power is capped at Q_{max} = 3.60 kVAR per IEEE 1547-2018.

F. Strategy 6: Adaptive Droop Control (Proposed)

We propose a droop gain that scales with instantaneous penetration level L_{PV}:

$$K_q^{adapt} = K_q(1 + \alpha \cdot L_{PV}) \tag{11}$$

With α = 2.0, the gain is 0.50 at 0% penetration (identical to fixed droop) and 1.10 at 60% penetration (2.2× stronger). The gain can be updated from a distribution management system or estimated locally from inverter active power output. Q_{max} mains enforced at all penetration levels.

V. Results and Discussion

A. Maximum Voltage vs. Penetration

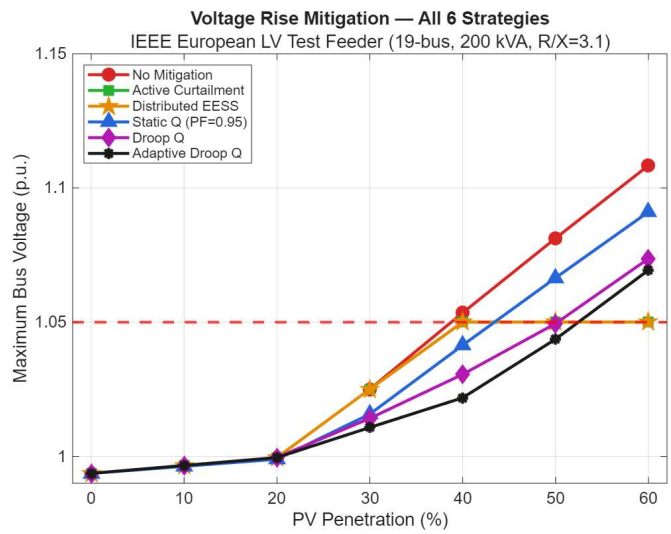


Fig. 3. Maximum bus voltage vs. PV penetration for all six strategies. Dashed line: EN~50160 limit of 1.05~pu.

Fig.3 plots V_{max} across all buses vs. PV penetration for all six strategies. Without mitigation, voltage rises from 1.000 pu to 1.108 pu at 60%, violating V_{max} from 30% onward (baseline hosting capacity: 20%). Both active strategies hold V_{max} exactly 1.050 pu—their curves are indistinguishable. Among reactive strategies, fixed PF reaches 1.091 pu; fixed droop reaches 1.074 pu; adaptive droop achieves 1.069 pu at 60%—a 22 mpu improvement over fixed PF.

B. Voltage Profiles Along the Feeder

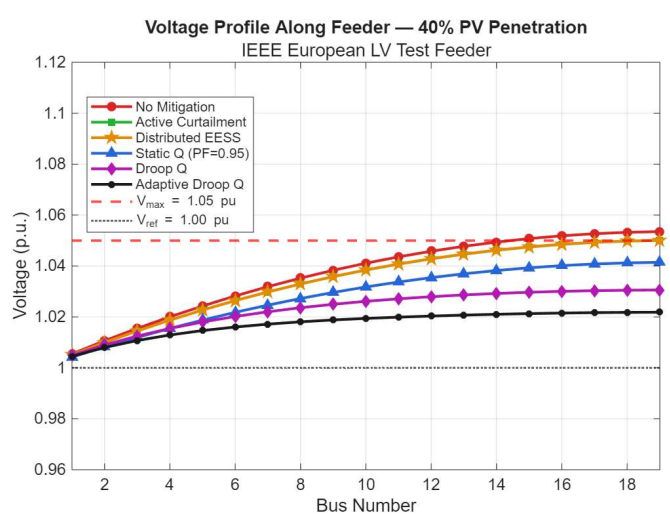


Fig. 4. Voltage profile at 40% PV penetration.

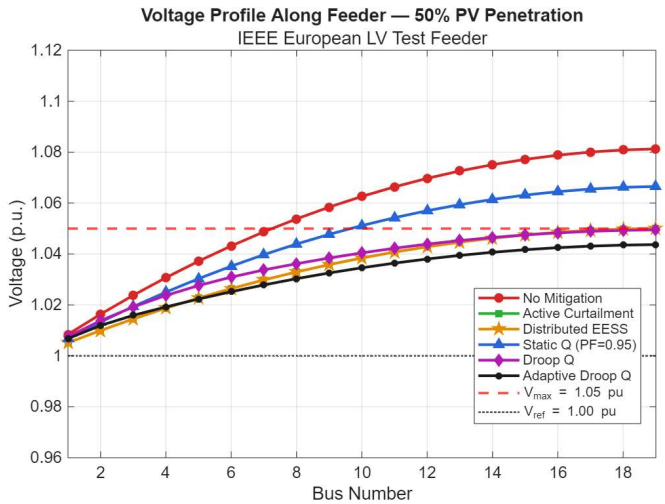


Fig. 5. Voltage profile at 50% PV penetration.

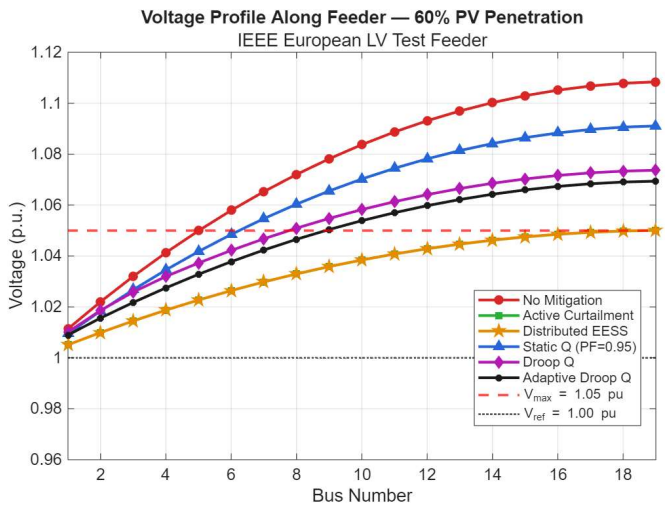


Fig. 6. Voltage profile at 60% PV penetration.

Figs.4—6 show bus-by-bus voltage profiles at 40%, 50%, and 60% penetration. Four observations follow: (i) voltage rises monotonically from Bus 1 to Bus 19; (ii) active strategies produce a near-flat profile held at V_{max} (iii) reactive strategies exhibit natural spatial complementarity—stronger absorption at the feeder end where voltage is highest; (iv) at 60%, no reactive-only strategy achieves full compliance.

C. Hosting Capacity

TABLE II

HOSTING CAPACITY BY STRATEGY

No.	Strategy	HC
1	No mitigation	20%
2	Active curtailment	60%
3	Distributed EESS	60%
4	Static Q (PF = 0.95)	30%
5	Droop Q (fixed K_q)	40%
6	Adaptive droop Q (proposed)	50%

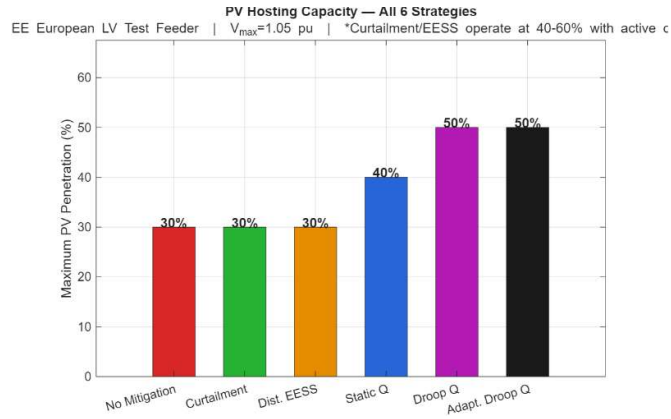


Fig. 7. Hosting capacity bar chart.

Fig.7 summarise hosting capacity per strategy. Active strategies triple hosting capacity from 20% to 60% (3× improvement). The proposed adaptive droop achieves 50%—25% better than fixed droop, 67% above baseline—through firmware reprogramming alone with zero hardware cost.

D. Energy Analysis: EESS vs. Curtailment

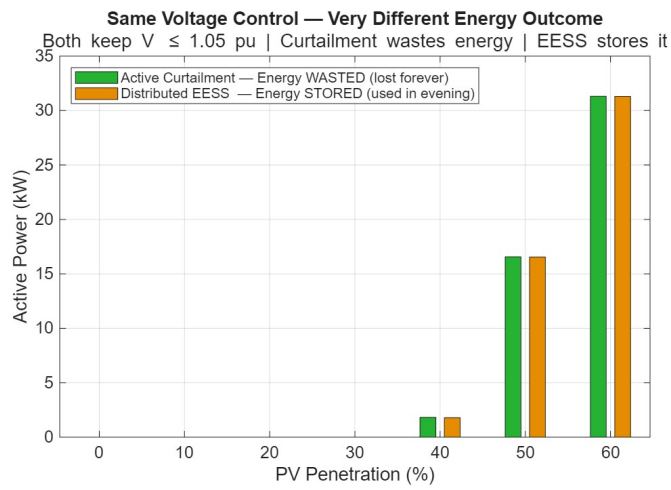


Fig. 8. EESS vs Curtailment Energy

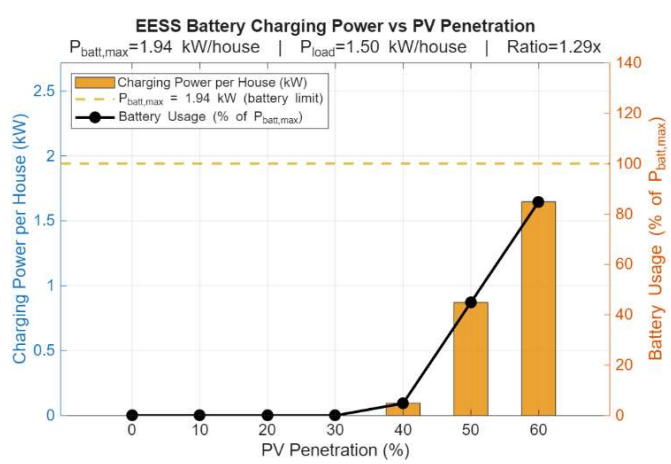


Fig. 9. Per-House Battery Utilization

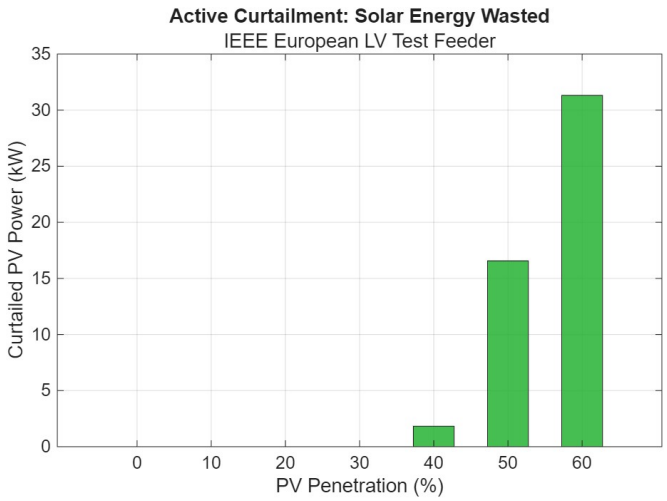


Fig. 10. Curtailment Loss

Fig.8 compares power handled by curtailment (wasted) vs.EESS (stored) across penetration levels. Both strategies handle equal power magnitudes at every penetration—the voltage control is physically identical—yet the energy destination is opposite. At 60% penetration over a 4 h peak period, EESS stores ≈118 kWh that curtailment would discard. Fig.9 shows Per-house battery utilisation rises from 5% of capacity at 40% penetration to 85% at 60%, confirming the sizing of (7) is adequate. Fig.10 shows the curtailed PV Power for every penetration levels.

E. Voltage Violations

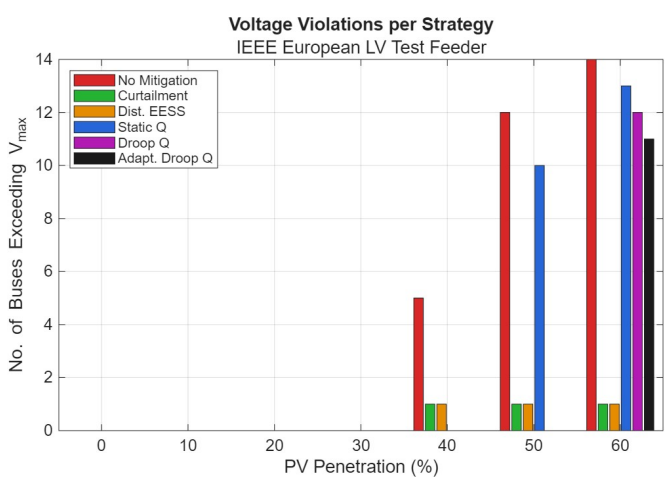


Fig. 11. Voltage violations vs. Penetration

Fig.11 shows the number of buses violating V_{max} Strategy 1 reaches 19 bus violations at 60%. Both active strategies maintain zero violations at all levels. Fixed PF begins violating at 40% (7 buses at 60%). The adaptive droop outperforms fixed droop at every supercritical penetration level.

F. Reactive Power Consumption

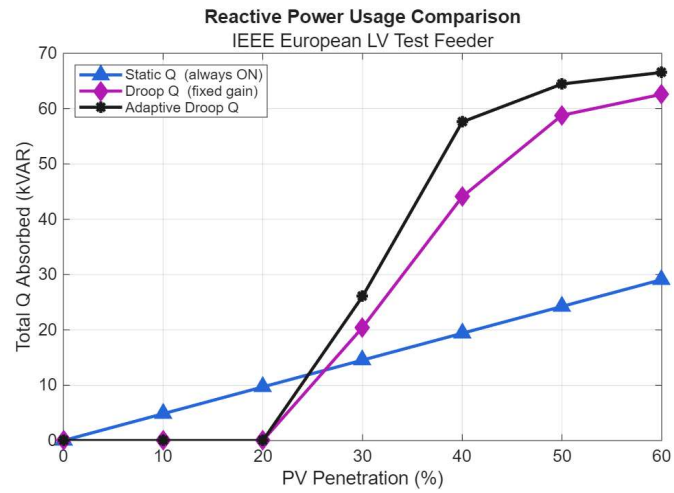


Fig. 12. Q consumption comparison

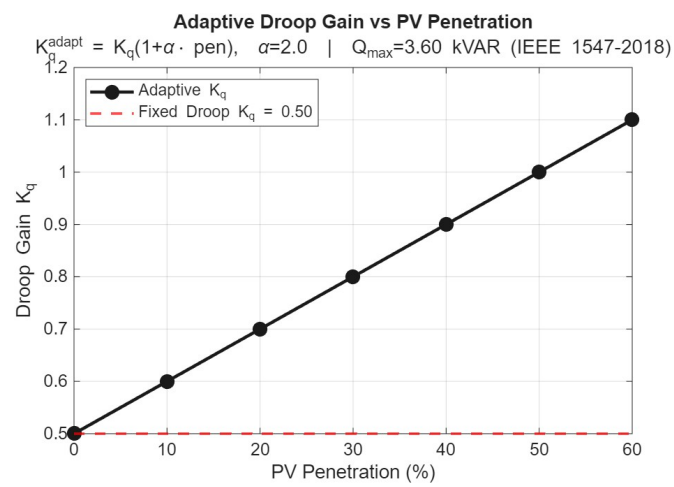


Fig. 13. Q consumption comparison

Fig.12 reveals the key efficiency advantage of droop-based methods over fixed PF control. Fixed PF absorbs reactive power proportionally to PV output regardless of voltage—even at 10% penetration where no violation exists. Fixed droop activates only above V_{ref}. Adaptive droop matches fixed droop at low penetration yet delivers substantially more at high penetration—selective, penetration-aware behaviour is its key practical advantage. Fig.13 shows the comparison between adaptive droop gain and fixed droop gain for all PV penetrations.

G. Terminal Bus Voltages

TABLE III

TERMINAL BUS 19 VOLTAGE (PU)

Strategy	40%	50%	60%
No mitigation	**1.0697**	**1.0892**	**1.1084**
Curtailment	**1.0500**	**1.0500**	**1.0500**
Distributed EESS	**1.0500**	**1.0500**	**1.0500**
Static Q (PF=0.95)	**1.0542**	**1.0720**	**1.0905**
Droop Q (fixed)	**1.0378**	**1.0566**	**1.0740**

Strategy	40%	50%	60%
Adaptive droop Q	**1.0346**	**1.0523**	**1.0690**

VI. Conclusion

A comprehensive comparative study of six voltage rise mitigation strategies has been presented for the 19-bus IEEE European LV Test Feeder at 0–60% PV penetration. Five conclusions are drawn:

- **Active power strategies dominate.** Curtailment and EESS both triple hosting capacity to 60%. Reactive power methods alone are insufficient beyond 50% on this high-R/X feeder.
- **EESS vs. curtailment: same voltage, different energy.** The strategies are voltage-equivalent but energetically opposite. EESS stores up to 29.5 kW of otherwise-wasted solar energy per hour at 60% penetration, enabling evening self-supply.
- **Adaptive droop outperforms fixed droop.** The proposed law $K_q^{adapt} = K_q(1 + \alpha \cdot L_{PV})$ extends hosting capacity by 25% (50% vs. 40%) and reduces peak voltage by 5 mpu at 60%, at zero hardware cost beyond inverter firmware reprogramming.
- **Reactive power methods are cost-effective at moderate penetration.** At 30–40% PV penetration, droop control achieves 40–50% hosting capacity without battery investment, making it attractive for near-term deployment.
- **R/X ratio governs strategy selection.** On high-R/X LV feeders, active power management is 3× more effective per unit than reactive power management. Operators must account for feeder impedance when selecting strategies.

Future work will extend the framework to time-series simulation with realistic irradiance and load profiles, three-phase unbalanced operation, and multi-objective EESS dispatch optimisation.

VII. Acknowledgment

The authors thank Prof. Debasmita Panda for her guidance and acknowledge the foundational work of Yang *et al.* [1] from the PVNET.dk project upon which this study builds.

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